Pradyumna Kumar Sahoo

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Summary

Applied AI Data Scientist with close to 5 years of experience deploying production AI/ML across healthcare and enterprise financial systems. Architected and scaled GraphRAG pipelines, multi-agent frameworks, and fine-tuned open-source LLMs running in live clinical workflows and rolled out organization-wide to all ADP employees. Experience leading engineers, domain experts, and academic researchers (IIT Madras, IIT Hyderabad) from data curation to production model serving.

Experience

Data Scientist Aug 2025 – Present
Mondee Pvt. Ltd., Hyderabad

- **Clinical Decision Support (GraphRAG):** Architected a GraphRAG audio chatbot enabling clinicians across Surekha Hospital Chain and BhaktiVedant Hospital to query alternative diagnoses and treatment paths; extracted structured medical entities (NER) into Neo4j and integrated retrieval via LangChain to eliminate unsupported answers through traceable source attribution.
- **Medication Safety Agent (MCP):** Developed an autonomous medication-safety agent integrated into the clinical chat interface; queried live RxNorm and PubMed data to detect drug–drug interactions, formulated patient-specific dosage regimens, and deployed the tool as an MCP service accessible to internal agent fleets.
- **Clinical LLM Fine-Tuning & Serving:** Coordinated with researchers from IIT Madras and IIT Hyderabad on dataset curation, training, and evaluation to fine-tune `MedGemma-27b-text-it` using Unsloth; served the model via vLLM for high-throughput clinical NLP inference.
- **Clinical Voice Agent Pipeline:** Spearheaded an end-to-end medical voice workflow by establishing live clinical audio collection, expert annotation, and quality control; fine-tuned `gemma3n-e2b-it` using LoRA and served it on TGI to transcribe consultations in real time and generate Doctor's Notes.

Senior Member Technical (AI/ML) Dec 2023 – Jul 2025
ADP India Pvt. Ltd., Hyderabad

- **Workplace Automation Assistant:** Built an autonomous workplace assistant using Google ADK and AWS OpenSearch that generates context-aware emails and schedules meetings against live calendars directly within chat; saved approximately 24 hours per user monthly across company-wide rollout to all ADP employees.
- **Knowledge Graph-Grounded Financial RAG:** Coupled RAG pipelines with an AWS Neptune knowledge graph and Amazon Bedrock using LangGraph to resolve ambiguous queries, increasing financial response accuracy, reducing hallucinations, and earning Runner-up at the 2024 ADP Global Hackathon.
- **Payroll Process Mining Chatbot:** Implemented a process-mining conversational agent using Databricks and Power Automate over millions of client records, automatically identifying payment bottlenecks and resolving issues across an average of 73 payroll cycles per client.
- **Multilingual PII Redaction Pipeline:** Fine-tuned Transformer and spaCy NER models across 10+ Indic scripts to identify and redact sensitive personal and financial identifiers (including tax IDs, national IDs, account numbers, and names) in payroll and HR documents to ensure regulatory compliance.

Junior Data Scientist Jun 2021 – Dec 2023
Claim Genius Pvt. Ltd., Remote

- **Automated Vehicle Damage Segmentation:** Engineered a vehicle-part segmentation pipeline using Detectron2 served via FastAPI, achieving 95% mAP and reducing damage assessment turnaround by 26%; integrated GradCAM++ heatmaps to provide transparent model interpretability for claims reviewers.
- **Inference Monitoring & Root-Cause Tracing:** Built an MLflow-based inference monitoring system tracking test-time feature distributions, reducing model diagnosis and root-cause analysis time by 30%.
- **Super-Resolution Image Preprocessing:** Mitigated production errors from compressed and low-quality photos by integrating a SWIN2SR and NAFNet super-resolution and denoising ensemble, reducing model failures by 16% and improving downstream task accuracy by 12%.

Technical Skills

- **Generative AI and Agents:** LLM fine-tuning (full, PEFT, LoRA, QLoRA), instruction tuning, RLHF/DPO, RAG and GraphRAG, multi-agent orchestration, tool use / function calling, Model Context Protocol (MCP), LangChain, LangGraph, Google Agent Development Kit (ADK), DSPy, LiveKit.
- **ML Infrastructure and Serving:** vLLM, TGI, quantization (GPTQ, AWQ, bitsandbytes), mixed precision (bf16/fp16), gradient checkpointing, Docker, FastAPI, CI/CD for ML, MLflow, Weights & Biases.
- **ML and Deep Learning:** Python, PyTorch, TensorFlow, Scikit-learn, XGBoost, LightGBM, HuggingFace Transformers, Detectron2, OpenCV, spaCy; NLP, speech/ASR, computer vision, Graph Neural Networks (GNNs).
- **Knowledge Graphs and Data:** Neo4j, AWS Neptune, Qdrant, LanceDB, AWS OpenSearch, SQL, PostgreSQL, MongoDB, SQLite, Apache Spark (Databricks), Apache Airflow, ETL design.
- **Cloud Platforms:** AWS (Bedrock, Neptune, S3, Lambda, EC2, OpenSearch); GCP Vertex AI and Azure ML.
- **Languages:** Python, SQL, MATLAB, Bash.

Research Projects (TCS Big Data Lab)

- **Multi-Label Classification on Imbalanced Data:** Combined MLSMOTE with an LLSF-DL deep learning hybrid (Python, MATLAB, TensorFlow) to synthesize representations for rare tail labels, improving multi-label classification performance.
- **Session-Based Recommendation:** Designed a PyTorch Graph Neural Network (GNN) engine modeling structural graph dependencies to generate next-item recommendations during active user sessions.

Certifications

- Agentic Knowledge Graph Construction — DeepLearning.AI × Neo4j Aug 2025
- Building AI Voice Agents for Production — DeepLearning.AI × LiveKit Jul 2025
- Neo4j Fundamentals — Neo4j GraphAcademy Jul 2025
- Pretraining LLMs — DeepLearning.AI × Upstage Feb 2025

Education

- **M.Sc. Computer Science (Big Data Analytics)**, Central University of Rajasthan, Kishangarh